

Ch.2 Backing up

March 11, 2026

Game Forms

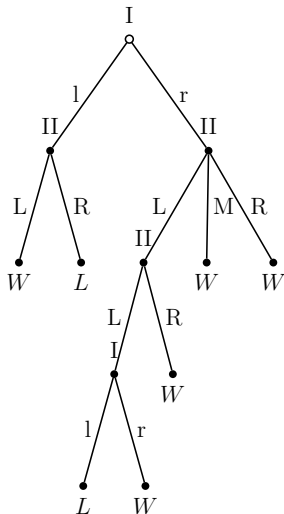
A Winning Strategy

Nim

A Strictly Competitive Game

Saddle Points

Consider a 2-person win-or-loss game with perfect information G :
 $W \succ_I L$, $L \succ_{II} W$.

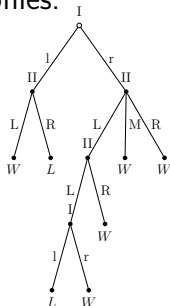


- ▶ In the previous slide, we have the *game tree* of G , or the *extensive form* of G .
- ▶ We could also present G 's *strategic form*.
 - ▶ Def: Strategy of i is a statement that specifies an action at each of i 's decision node.
 - ▶ Player I has 2 decision nodes, and I has 4 pure strategies: ll , lr , rl , rr .
 - ▶ How many pure strategies does player II have?

Strategic Form/Normal Form

We'll list 2 players' pure strategies in a matrix form and fill in the results of all strategy profiles.

		II		
		LLL	LLR	
I	ll	W		
	lr	W		
	rl	L		
	rr	W		



Solution Concepts

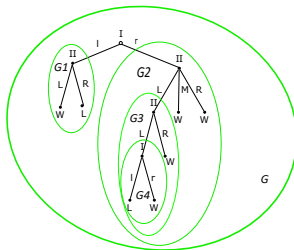
- ▶ extensive form: subgame perfect equilibrium (SPE)
- ▶ strategic form: Nash equilibrium (NE)
- ▶ An extensive form reveals more information about the game than a strategic form.
 - ⇒ More requirements can be imposed on players' choices in an extensive form than in a strategic form.
 - ⇒ $\{\text{SPE}\} \subseteq \{\text{NE}\}$

Solution Concept: A Winning Strategy

- ▶ Def: If i has a strategy for game H that wins H whatever strategy i 's opponent may use, then i 's strategy is called his winning strategy.
- ▶ If a player has a winning strategy, he'll surely adopt it.
- ▶ Existence of a winning strategy
 - ▶ Zermelo developed an algorithm in 1912 to analyze Chess, and confirmed the existence.
- ▶ Who has a winning strategy?

Subgame and Value

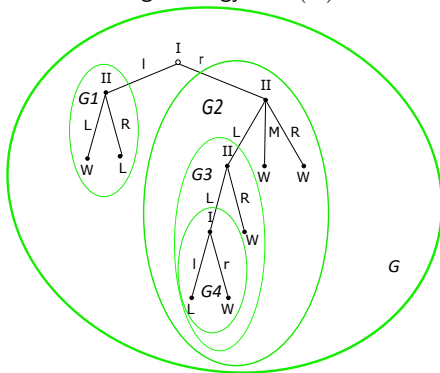
- ▶ Def: A subgame consists of a node x together with all of the game tree that follows x .
- ▶ 5 subgames of G :



- ▶ Def: Value of the subgame H , $v(H) = W$, if I has a winning strategy for H , and $v(H) = L$, if II has a winning strategy for H .

Subgame and Value

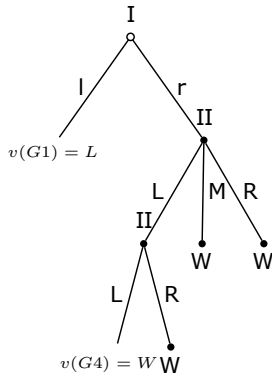
- Existence of a winning strategy $\Leftrightarrow v(G)$ well-defined.



- $v(G) = ?$, but ready answers when there remains only one decision node:
 $v(G1) = L$, $v(G4) = W$

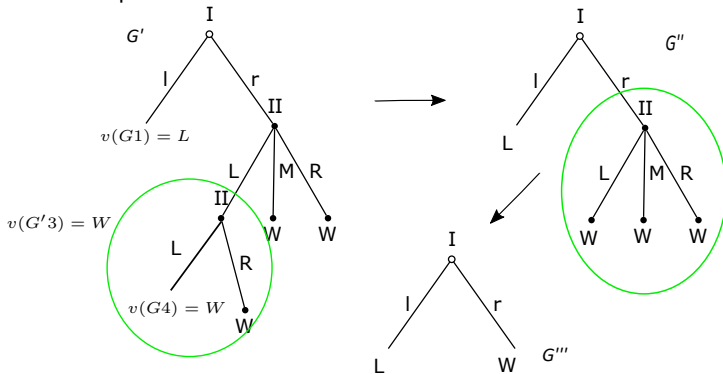
A Simplified Game, G'

- Game G' obtained from G by replacing $G1$ and $G4$ with their values



- ▶ Claim: If $v(G') = W$, then $v(G) = W$.
Proof: If $v(G') = W$, player I has a winning strategy s' for G' , i.e. player I can always direct the game to end w. a node labeled W . If the node corresponds to a subgame G_x of G , then $v(G_x) = W$, and player I has a winning strategy s_x in G_x .
The winning strategy s for player I in G consists of playing s' all the way until the subgame G_x is reached, then playing s_x . $\square$
- ▶ Similarly, if player II has a winning strategy in G' ($v(G') = L$), then player II has a winning strategy in G ($v(G) = L$).
- ▶ In sum, if $v(G')$ is well defined, so is $v(G)$ and $v(G) = v(G')$.

- ▶ $v(G')$ well defined?
- ▶ further simplifications



- ▶ $W = v(G''') = V(G'') = v(G') = v(G)$
- ▶ A 2-person finite win-or-loss game with perfect information has a value.

Who has the winning strategy?

- ▶ Nim
 - ▶ There are several piles of matchsticks.
 - ▶ Two players alternate in moving.
 - ▶ When it is your turn to move, you must select one of the piles and remove at least one matchstick from that pile.
 - ▶ The last player to take a matchstick is the winner.
- ▶ Nim is a 2-person finite win-or-loss game with perfect information.
- ▶ Nim has a value.

- ▶ 3 piles of matchsticks: 5, 9, 4 sticks in each pile
- ▶ Who has the winning strategy?

- ▶ a binary representation of the game:

	8	4	2	1
5	0	1	0	1
9	1	0	0	1
4	0	1	0	0

- ▶ Def: Nim is *balanced* if each column has an even number of 1s and *unbalanced* otherwise.
- ▶ The example is unbalanced.

Claim: A player that starts with an unbalanced Nim has a winning strategy.

Proof: (a) A player that starts with a balanced Nim cannot win immediately.

(b) $\forall$ balanced Nim, every move converts the game back to an unbalanced one.

(c) $\forall$ unbalanced Nim, there is a move to convert it to a balanced one.

Reconsider the example:

	8	4	2	1
5	0	1	0	1
9	1	0	0	1
4	0	1	0	0

The starting player should take 8 sticks from the 2nd pile. $\square$

Chess

- ▶ It is not a win-or-loss game. It could end with a draw D .

$$W \succ_I D \succ_I L$$

$$W \prec_{II} D \prec_{II} L$$

- ▶ Def: A 2-person game which has k different outcomes, $u_1, \dots, u_k$, is *strictly competitive*, if

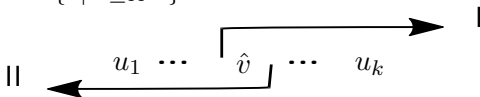
$$u_k \succ_I \dots \succ_I u_1$$

$$u_k \prec_{II} \dots \prec_{II} u_1$$

- ▶ Chess is strictly competitive.

Value of a strictly competitive game G

- Def: $v(G) = \hat{v}$, if two players can force an outcome no worse than $\hat{v}$ to themselves at the same time, i.e. player I can force an outcome in the set $W = \{u | u \succeq_I \hat{v}\}$ and player II can force an outcome in the set $L = \{u | u \succeq_{II} \hat{v}\}$.



- This extends the definition of the value in a win-loss game:



$v(G)$ exists

Lemma: Let T be any set of outcomes in a finite 2-person game with perfect information. Either player I can force an outcome in T , or player II can force an outcome in $\sim T$.

Proof: Relabel all outcomes in T with W , and all outcomes in $\sim T$ with L . Then it reduces to showing that any finite win-or-loss game has a value. $\square$

Theorem: Any 2-person finite strictly competitive game with perfect information w/o chance moves has a value.

Proof: Consider a game, G , with k different outcomes and $u_k \succ_I \dots \succ_I u_1$. Define W_{u_i} as follows:

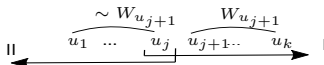
$$\begin{aligned} W_{u_1} &= \{u_1, \dots, u_k\} \\ W_{u_2} &= \{u_2, \dots, u_k\} \\ &\dots \\ W_{u_k} &= \{u_k\} \end{aligned}$$

Player I can force an outcome in W_{u_1} .

Let W_{u_j} be the smallest set in which player I can force an outcome.

If $j = k$, $v(G) = u_k$. If $j \neq k$, then player I cannot force an outcome in $W_{u_{j+1}}$.

From the lemma, player II can force an outcome in $\sim W_{u_{j+1}}$



$v(G) = u_j$. $\square$

Corollary: Chess has a value.

Saddle Points

- ▶ The concept of value of the game was developed earlier than the concept of NE and SPE.
- ▶ They are different:
 - ▶ Value is about the outcome.
 - ▶ Equilibrium is about the strategy profile.
- ▶ But are they related?
 - ▶ We'll define a saddle point and find its relation with the value and NE, hence connect the value and NE.

- ▶ Def: A strategy pair (s, t) is a saddle point of the strategic form of a strictly competitive game if, for I, the outcome v of (s, t) is no worse than any outcome in column t and no better than any outcome in row s .

	t	
	$\sqsubseteq_1 v$	
	$:$	
s	$\sqsubseteq_1 v \quad \dots \quad v \quad \dots \quad \sqsubseteq_1 v$	
	$:$	
	$\sqsubseteq_1 v$	

- Corollary: The strategic form of a finite, strictly competitive game of perfect information w/o chance moves has a saddle point (s, t) .

Proof: $\therefore$ value exists.

- ▶ (s, t) is a NE $\Leftrightarrow (s, t)$ is a saddle point.